

Differentiation Trajectory Analysis of Mouse Bone Marrow Hematopoietic Progenit...

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Report Title

**Differentiation Trajectory Analysis of Mouse
Bone Marrow Hematopoietic Progenit...**

Conversation Name

**Please perform differentiation trajectory
analysis on the attached hematopoieti...**

Report Time

2026-09-03 00:15

1. Results

1.1 Differentiation Trajectory Analysis of Mouse Bone Marrow Hematopoietic Progenitors

1.2 Overview

This report presents a complete single-cell differentiation trajectory analysis of mouse bone marrow hematopoietic progenitor cells (GSE72857, Paul et al. 2015). The analysis covers quality control, normalization and PCA, graph-based clustering, PAGA-based trajectory construction with diffusion pseudotime, branch point and cell fate identification, dynamic gene detection along pseudotime, a pseudotime-ordered heatmap of dynamic genes, and a summary of the main differentiation paths with key regulatory genes at each branch. All analyses were performed on 8,399 QC-passing cells.

1.3 Quality Control Filtering

The 27,297-gene x 10,368-cell expression matrix was loaded and per-cell total UMI and detected gene counts were computed. Design metadata was merged by Well_ID key. Cells were filtered to retain those with total UMI ≥ 500 AND detected genes ≥ 200 .

Passing cell composition by sorted population: Unsorted myeloid 2,732; Cebpa KO 1,384; CMP Flt3+ Csf1r+ 1,132; CMP Irf8-GFP+ MHCII+ 1,116; Cebpa control 890; Cebpe KO 559; CMP CD41 336; Cebpe control 250. All 108 empty wells (Number_of_cells=0) were correctly excluded.

1.4 Normalization and Dimensionality Reduction

Log-normalization was applied to the 8,399 QC-passing cells: counts were divided by per-cell total UMI, multiplied by 10,000, and log1p-transformed. The top 2,000 highly variable genes were selected using Seurat-style mean-binned standardized variance (top hits: Cd74, H2-Aa, Prtn3). PCA was computed on the z-scored HVG matrix via full SVD, retaining the top 50 PCs as the cell embedding.

Metric	Value	Notes
n_genes	27297	
n_cells	8399	
n_hvg	2000	
n_pcs	50	
pca_pc1_evr	0.01655740685231714	
pca_cumulative_evr_50pcs	0.11435249005275092	

1.5 KNN Graph Construction and Clustering

A KNN graph was built on the 50-PC embedding (k=30, method='umap', random_state=0). Leiden clustering at resolution 1.0 produced 15 clusters with graph modularity 0.8012. A 2-D UMAP embedding was computed for visualization.

Cluster sizes ranged from 12 to 1,178 cells. The largest clusters were 1 (1,178), 8 (1,056), 7 (887), 5 (876), and 4 (832). The smallest clusters were 3 (61 cells) and 11 (12 cells), likely rare transition states or low-complexity outliers.

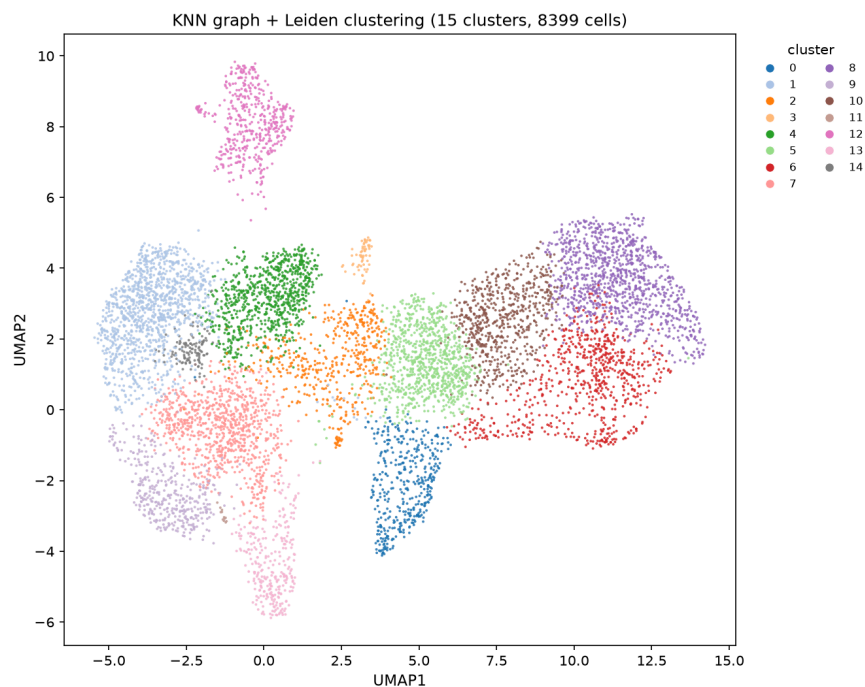


Figure 1 UMAP embedding of 8,399 QC-passing cells colored by Leiden cluster (15 clusters, resolution 1.0).

The UMAP shows a continuous manifold of hematopoietic progenitor states with multiple major branches, consistent with a differentiation trajectory. Cluster 4 (root) sits at a central position with branches extending toward distinct lineage fates.

1.6 Trajectory Construction (PAGA + Diffusion Pseudotime)

A differentiation trajectory was constructed using PAGA (model v1.2) on the KNN graph, with a root-anchored diffusion pseudotime (Haghverdi et al. 2016 formulation). The root was inferred data-driven as Leiden cluster 4 (832 cells) using a primitive score based on CMP-stage stemness markers (Kit, Cd34, Gata2, Mecom, Procr, Hlf, Mpl) minus committed-lineage module scores, plus a sorted-gate metadata prior. The root cell was W60016.

The PAGA MST rooted at cluster 4 yielded 6 lineages to terminal clusters: L1->c6 [4,2,5,10,8,6] (4,579 cells), L2->c0 [4,2,5,0] (2,603), L3->c9 [4,2,7,9] (2,590), L4->c13 [4,2,7,13] (2,486), L5->c1 [4,14,1] (2,135), L6->c12 [4,2,3,12] (1,841). Per-lineage principal curves (120 nodes each) were fit in PCA space.

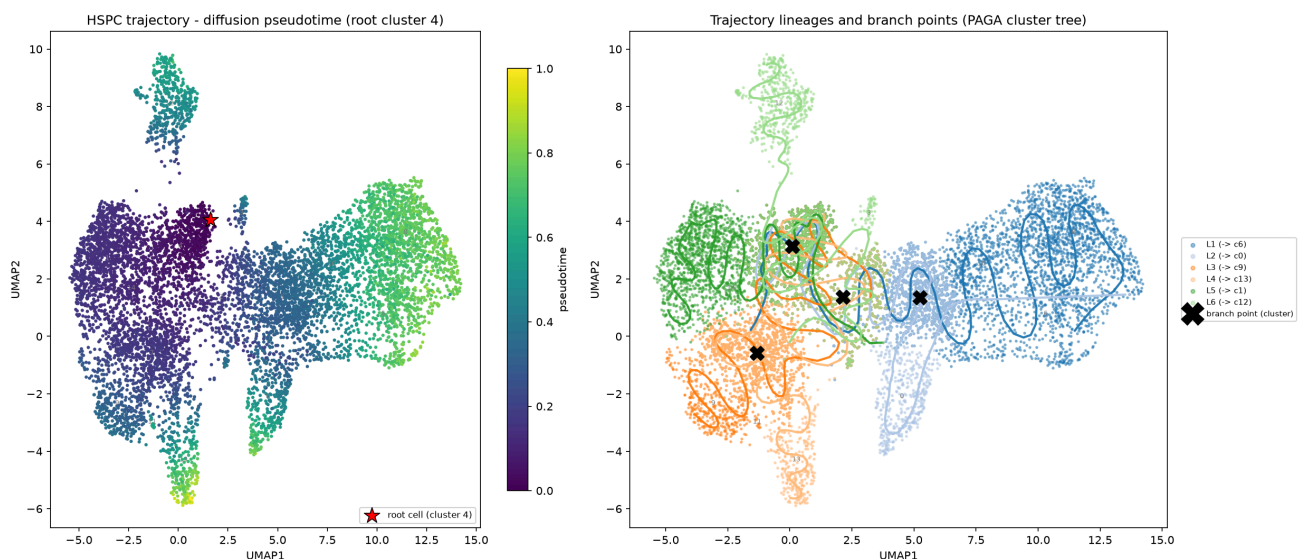


Figure 2 Trajectory with pseudotime coloring (left) and lineages with branch-point markers (right).

Diffusion pseudotime increases from the root cluster 4 (median pseudotime 0.047) outward to terminal states. Six lineages emanate from the root through branch points at clusters 2, 5, and 7, capturing the major myeloid differentiation paths.

1.7 Branch Points and Cell Fate Decisions

Four branch points were identified on the rooted PAGA-MST tree (nodes with ≥ 2 downstream branches), ordered by tree depth and median pseudotime. Terminal cluster fates were annotated using curated marker modules (MonoDC, Granulocyte, MegE, EosBaso, stem_CMP) with gene-z scoring across the 6 terminal clusters.

Branch point	Cluster	Median pseudotime	Branches	Branch fractions	Fate decisions
BP1	c4 (root)	0.047	2	82.8% / 17.2%	MegE trunk vs MonoDC
BP2	c2	0.190	3	9.2% / 63.7% / 27.1%	MonoDC vs MegE vs EosBaso/Granulocyte trunk
BP3	c7	0.167	2	57.6% / 42.4%	EosBaso vs Granulocyte
BP4	c5	0.325	2	14.8% / 85.2%	Megakaryocyte (c0) vs Erythroid (c6)

Terminal cluster fates: c0 MegE-megakaryocyte (Pfl/Irga2b; CMP CD41 gate), c1 MonoDC-early (CMP Flt3+ Csf1r+ gate; GMP-like mixed state), c6 MegE-erythroid (Klf1/Gata1/Gypa), c9 EosBaso (Cebpe+), c12 MonoDC (H2-Aa/Cd74; Irf8-GFP+ MHCII+ gate), c13 Granulocyte (Prtn3/Elane/Mpo/Ctsg; Cebpe-KO dominated).

Differentiation trajectory branch points and cell fate decisions (GSE72857, 8,399 cells)

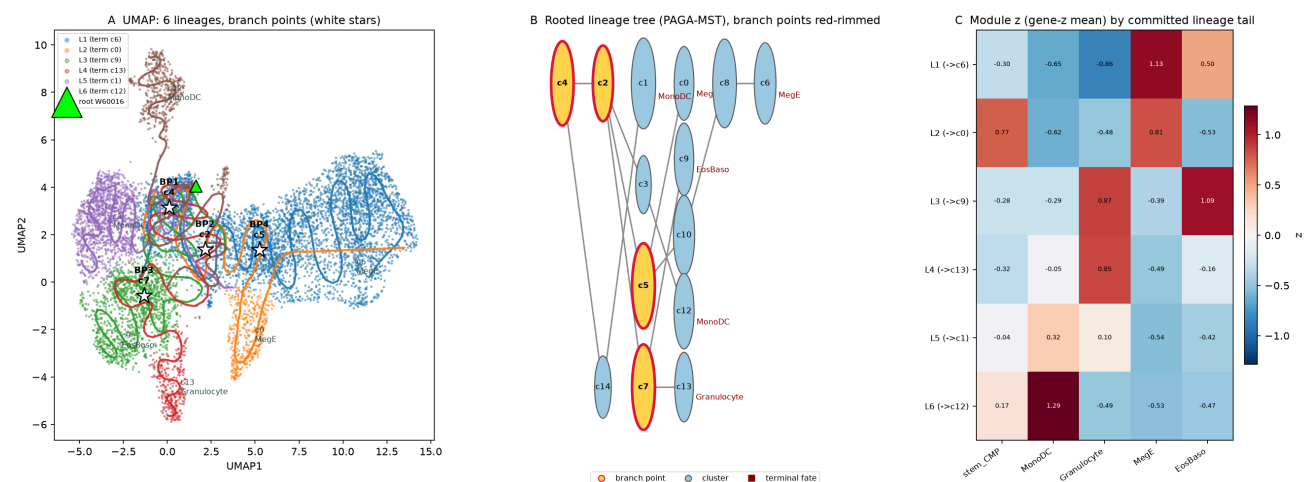


Figure 3 Branch point analysis: UMAP with lineages and branch stars (A), rooted lineage tree with branch points and terminal fates (B), and lineage-module z heatmap (C).

Four branch points (BP1-BP4) were identified. The rooted tree shows the hierarchical fate decisions: first a MegE vs MonoDC split at the root, then a three-way split (MonoDC/MegE/EosBaso-Granulocyte), followed by EosBaso vs Granulocyte and megakaryocyte vs erythroid splits.

1.8 Dynamic Genes Along Pseudotime

Dynamic genes were identified using a Gaussian GAM of log-normalized expression versus global pseudotime (cubic B-spline basis, $p=8$, clamped quantile knots), with an F-test against an intercept-only null and Benjamini-Hochberg FDR correction. Of 12,528 genes tested (detected in ≥ 10 cells, variance > 0), 10,415 were dynamic at $\text{padj} < 0.05$.

Biological sanity checks confirmed expected patterns: stem/multipotent markers down-regulated at the root (Cd34, Kit, Mecom, Hlf, Mpl); early-priming genes early-up (Flt3, Gata2, Cebpe); late fate genes late-up (erythroid Klf1/Gata1; granulocyte Prtn3/Elane/Mpo/Ctsg; Ly6c2, Cd74). Top genes by $r_squared$ (Rh�, H2afy, Mt2, Sphk1, Fam132a, Blvr, Ermap, Car2, Cpox) are enriched in erythroid/maturation programs.

1.9 Pseudotime-Ordered Heatmap of Dynamic Genes

A heatmap was generated with cells ordered by global pseudotime ascending (root cell W60016 at pseudotime 0.0 first, W53729 at pseudotime 1.0 last). The top 50 dynamic genes per expression pattern (down_regulated, early_up, transient, late_up) by $r_squared$ were plotted (200 genes total), with pattern blocks in the order down_regulated $\rightarrow$ early_up $\rightarrow$ transient $\rightarrow$ late_up. Per-gene z-scores of log-normalized expression were clipped to $[-3, 3]$ with an RdBu_r colormap.

The plotted subset includes key markers Cd34 (down_regulated), Flt3, Gata2 (early_up), and Klf1, Prtn3, Mpo (late_up). A supporting overview figure renders all 10,415 dynamic genes. The exact plotting data (z-scores, pseudotime order, gene pattern) is stored in heatmap_data.csv.

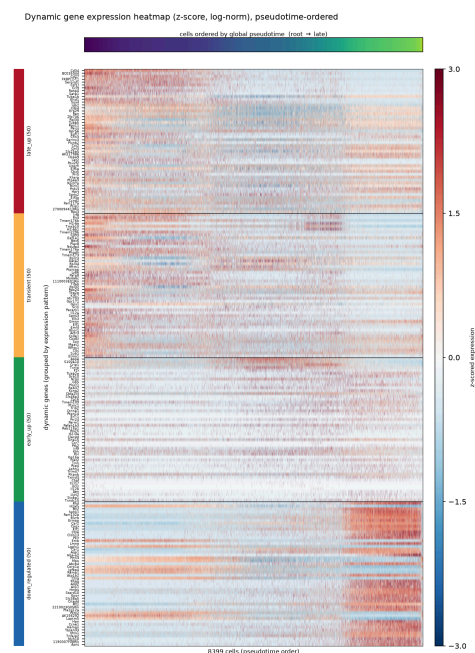


Figure 4 Heatmap of 200 dynamic genes (top 50 per expression pattern by $r_squared$) ordered by pseudotime, with pattern blocks down_regulated, early_up, transient, late_up.

The heatmap shows clear blocks of coordinated gene expression changes along pseudotime. Down-regulated genes (e.g., Cd34, Kit) are active at the root and decline; early-up genes (e.g., Flt3, Gata2) peak in the first third; late-up genes (e.g., Klf1, Prtn3, Mpo) rise toward terminal fates, reflecting lineage-specific maturation programs.

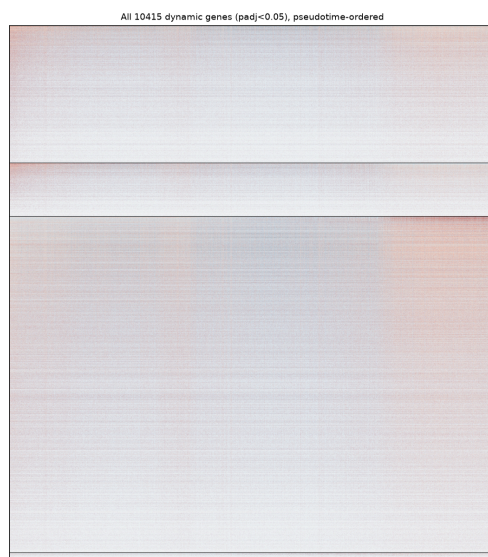


Figure 5 Supporting overview heatmap of all 10,415 dynamic genes ordered by pseudotime.

The full dynamic gene set confirms the pattern structure visible in the main figure, with large blocks of late-up genes dominating the late pseudotime columns as cells commit to terminal lineages.

1.10 Main Differentiation Paths and Key Regulatory Genes

Six main differentiation paths were summarized, each with key regulatory transcription factors and marker genes (derived from dynamic genes enriched in the path tail vs all other cells, with TF calls from a curated 98-gene mouse hematopoiesis regulator panel).

Path	Cluster path	Fate	Key regulatory TFs	Key markers
L1	c4->c2->c5->c10->c8->c6	MegE-erythroid	Klf1, Gata1, Gfi1b, Zfpm1, Tal1, Mllt3	Rhd, Ermap, Sphk1, Pklr, Ank1
L2	c4->c2->c5->c0	Megakaryocyte (MegE)	Pbx1, Fli1, Etv6, Runx1, Gata2	Pf4, Itga2b, Vwf, Gp9, Cd9
L3	c4->c2->c7->c9	EosBaso	Cebpe, Gfi1, Cebpa	Prg2, Epx (eosinophil program)
L4	c4->c2->c7->c13	Granulocyte	Cebpa, Gfi1	S100a8, S100a9, Lcn2, Chi3l3
L5	c4->c14->c1	MonoDC-early (GMP-like)	Flt3, Irf8, Cd34	Mpo, Prtn3, Ctsg (mixed)
L6	c4->c2->c3->c12	MonoDC	Irf8, Id2, Batf3, Flt3	H2-Aa, H2-Eb1, Cd74, Aif1

Key regulatory genes at each branch point: BP1 (root) B1 MegE trunk (Klf1/Gata1/Gata2) vs B2 MonoDC (Flt3/Irf8); BP2 B1 MonoDC (Id2/Batf3/Irf8), B2 MegE (Gata1/Klf1/Zfpm1/Gfi1b/Tal1/Stat5a), B3 EosBaso-Granulocyte trunk (Gfi1/Cebpa/Cebpe/Fli1); BP3 B1 EosBaso (Cebpe/Cd34/Gfi1/Klf1) vs B2 Granulocyte (Id2/Irf8 + S100a8/S100a9/Lcn2); BP4 B1 megakaryocyte (Pbx1/Fli1/Mef2c/Etv6/Runx1/Itga2b/Pf4/Vwf) vs B2 erythroid (Klf1/Mllt3/Tcf3/Gata1/Ermap/Rhd).

最终答案：Differentiation trajectory analysis of 8,399 QC-passing mouse bone marrow hematopoietic progenitor cells (GSE72857) identified a PAGA-based trajectory with 6 lineages, 4 branch points, and 10,415 dynamic genes (padj<0.05). The root (cluster 4, 832 cells) is a CMP-stage multipotent state. The trajectory captures the major myeloid fate decisions: (1) an early MegE vs MonoDC split at the root (BP1); (2) a three-way split at cluster 2 (BP2) into MonoDC, MegE, and an EosBaso/Granulocyte trunk; (3) an EosBaso vs Granulocyte split at cluster 7 (BP3); and (4) a megakaryocyte vs erythroid split at cluster 5 (BP4). Key regulatory genes include Klf1/Gata1/Gfi1b (erythroid), Pbx1/Fli1/Runx1 (megakaryocyte), Cebpe/Gfi1 (eosinophil/basophil), Cebpa/Gfi1 (granulocyte), and Irf8/Id2/Batf3 (monocyte/DC). The pseudotime-ordered heatmap of 200 top dynamic genes (50 per pattern) shows clear down-regulated, early-up, transient, and late-up expression blocks reflecting the differentiation program.